# Compartment-resolved smFISH quantification in iPSC-derived neurons

Technical documentation of the complete pipeline: what every step does, why it is implemented this way, how it relates to established smFISH analysis, and how it was validated. pipeline version 4 (October 2026). Practical instructions are in the seven HowTo files.

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## 1. Scope and overview

The pipeline quantifies single-molecule FISH (smFISH) spots in human iPSC-derived glutamatergic (Ngn2-induced) neurons imaged by spinning-disk confocal microscopy. Each detected spot is fitted, assigned to the soma, the dendrites of the imaged neuron or the surrounding axonal network, filtered, and counted as a density per compartment area. Statistics compare control vs protease (a protocol step improving probe access) and DIV9 vs DIV27 for six probes, three differentiations and two timepoints (~1,300 images on four camera setups). The analysis is two-dimensional, on maximum-intensity projections (MIPs). Spot coordinates are detected upstream (napari/Spotiflow; Dominguez Mantes et al., 2025).




*Figure 1. Scripts and order of use. Grey: optional steps.*

## 2. Data, imaging setups and pixel calibration

All physical quantities depend on the pixel size. Four acquisition setups exist; laser and camera hardware changed between sessions, and samples that failed were redone later from other differentiations. Control and protease of a pair were always imaged in the same session on the same setup.

| Setup label | Image size | Pixel | Notes |
|---|---|---|---|
| 2304px_0.065um | 2304 × 2304 | 0.065 µm | most of replicates 2 and 3 |
| 2048px_0.065um | 2048 × 2048 | 0.065 µm | three DIV9 pairs |
| 1024px_0.130um | 1024 × 1024 | 0.13 µm | Nikon/Andor; part of DIV9 replicate 1, one DIV9 pair of replicate 2 |
| 1024px_0.130um_VisiView | 1024 × 1024 | 0.13 µm* | VisiView/EMCCD; all of DIV27 replicate 1, two DIV9 pairs. *metadata unit "cm" is wrong; 0.13 µm/px set by the script |

**Independent verification.** A Gaussian approximation of a high-NA PSF has σ ≈ 0.21 λ/NA (Zhang et al., 2007), ≈ 100 nm for Cy5 emission. Fitted single spots had σ = 1.55-1.75 px at 0.065 µm/px (101-114 nm) and σ ≈ 1.0 px at 0.13 µm/px (130 nm), consistent with the calibrations. The pixel size is read from every image, so mixed folders are handled. A wrong factor of 2 changes diameters 2× and densities 4× without any error message, which is why an early version (fixed 0.13 µm/px on 0.065 µm data) had to be replaced.

## 3. File discovery and pairing

Timepoint (div#), replicate (n=#), probe (name list) and condition ("prot" in the folder name) are parsed from the folder path. The sample ID is the text before the channel token (_C<n>_, _w<n>conf, _conf); a file belongs to a sample only if the ID appears as a whole unit (no letter or digit directly before or after). Zero or several matches → the image is skipped with a warning, never guessed. **Why:** a plain substring match with first-hit selection silently paired sample 1 with files of sample 10 in an earlier version. Images without a soma mask are ignored by design.

## 4. Spot detection

Detection is performed with Spotiflow (napari plugin, Hybiss model). Two properties of the data matter for every downstream step:

- **Two intensity populations.** Permissive detection yields a large dim population (often ~95% of candidates) and a bright population about 10× brighter. Dim puncta are 5-11× more frequent inside cells than in cell-free background in a tested image, consistent with off-target single-probe binding inside cells, the classic reason for intensity thresholds in smFISH (Raj et al., 2008). That some are partially labelled mRNAs cannot be excluded; a no-probe or non-targeting probe control would settle it.
- **Per-image thresholds.** In the original detections the Spotiflow probability threshold was set per image. In two images of one pair the effective cutoff differed ~2-fold (92% vs 31% of compact spots above 500 counts detected). Per-image thresholds chosen without blinding can shift comparisons; a single setting per setup is preferable.

`spotiflow_batch.py` (optional) implements a reproducible alternative: images are normalised with fixed values per control/protease pair (1st percentile to median + H × robust background noise), so control and protease are scaled identically; a calibration grid (model, 2× downsampling for 0.065 µm data, three H values) is evaluated on random calibration images plus at least one image per condition of every pair; the probability threshold is the highest at which every calibration image reaches ≥ 95% recall (median ≥ 98%) of bright reference spots (classical LoG detection + fit, bright population defined by a two-component mixture on log intensity). Spotiflow itself could not be run in the development environment; a first full trial produced overdetection, so this route is currently not used for the reported analysis.

## 5. Compartment masks

```
Soma     = manual soma label > 0 (never changed)
Dendrite = dilate(dendrite map > PROB_THRESHOLD, DENDRITE_DILATION_UM)  AND NOT soma
Axon     = dilate(neurite map  > PROB_THRESHOLD, AXON_DILATION_UM)      AND NOT dendrite AND
NOT soma
Background (diagnostics) = everything else
```

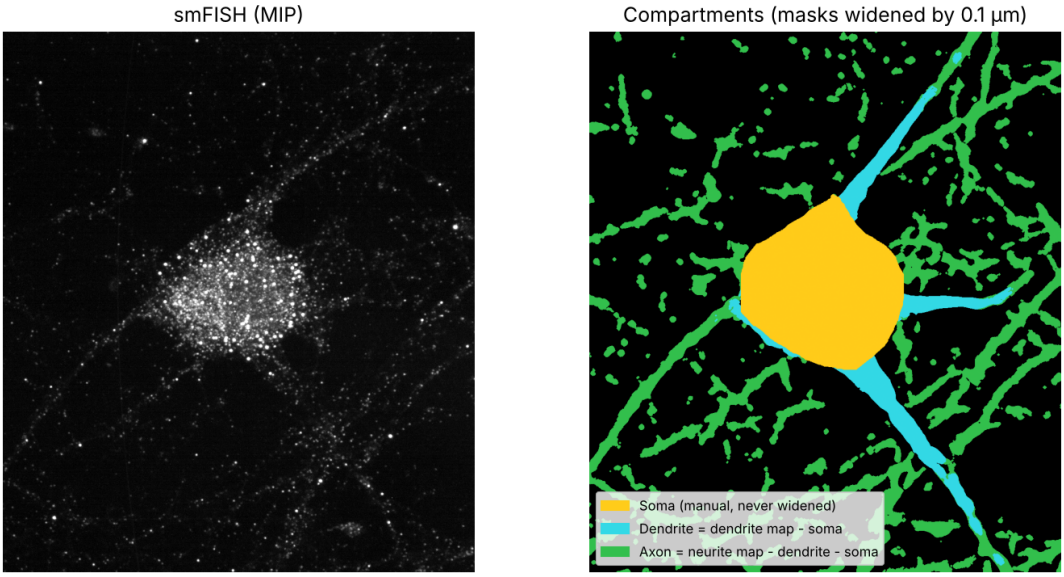



*Figure 2. Example image: smFISH MIP and the three mutually exclusive compartments (masks widened by 0.1 µm).*

**Mutually exclusive** masks count every spot once; priority follows annotation certainty (manual > targeted dendrite classifier > general neurite classifier). The axon compartment contains all non-dendritic neurites in the field, including axons of neighbouring neurons. Ilastik maps (Berg et al., 2019) are thresholded at PROB_THRESHOLD (0.35 in the current analysis).

**Dilation.** Ilastik masks are drawn tightly, so spots on neurite edges fall outside. Spot density in the rings added by successive widening steps (test image, spots per 100 µm²): inside 58.8; ring 0-0.1 µm 35.1; 0.1-0.2 µm 30.3; 0.2-0.3 µm 24.8; 0.3-0.4 µm 19.6; background 11.5. Every ring is richer than background, the first the most. Dilation works in whole pixels: 0.1 µm becomes 2 px at 0.065 µm/px and 1 px at 0.13 µm/px, i.e. exactly 0.13 µm on every setup (about one resolution element), whereas 0.2 µm would give 0.195 vs 0.26 µm. 0.1 µm is therefore recommended. Densities are spots per µm² of mask; area-based density is valid when the same classifier and threshold are applied to all conditions.

## 6. Spot fitting, background and noise

A 2D elliptical Gaussian plus constant offset is fitted by bounded least squares in a 7 × 7 px window around each detection (amplitude ≥ 0, 0.5 ≤ σ ≤ 5 px), the standard approach for smFISH spot measurement (Femino et al., 1998; Thompson et al., 2002; Mueller et al., 2013; Imbert et al., 2022).

- **Background** = fitted offset. The spot tail is part of the model, so it is not counted as background. An annulus 1.5-2.25 σ from the centre (used in v1) lies on the spot tail (32% of peak at 1.5 σ): on 300 spots it gave 353 vs 224 counts, a bias comparable to typical spot amplitudes (~160 counts).
- **Noise** = SD of the fit residuals with 6 fitted parameters subtracted; recovered simulated noise within ~5%.
- **SNR** = amplitude / noise. Spot height and noise scale together with exposure, laser power and gain, so the ratio transfers between settings, unlike absolute counts.
- **Integrated intensity** = $2\pi \cdot A \cdot \sigma x \cdot \sigma y$, the measure used to estimate molecules per spot (Mueller et al., 2013; Tsanov et al., 2016). A 7 vs 11 px window gave r = 0.96 for integrated intensity.

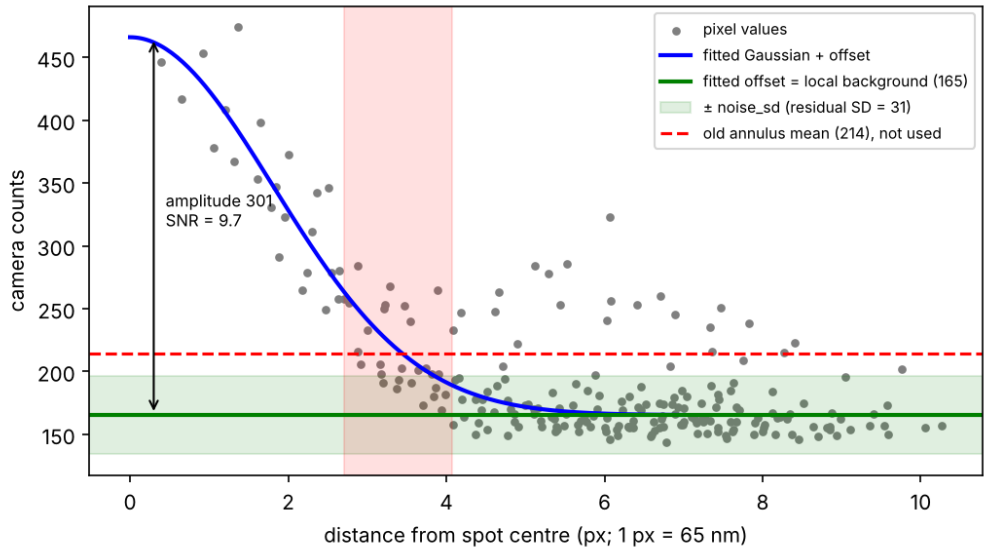



*Figure 3. Radial profile of a counted axonal spot: fitted model, background (offset), noise band and SNR. The annulus mean (red, not used) sits on the spot tail.*

## 7. Spot filters

One function applies identical filters to every compartment and to control and protease. The first failed criterion is stored as the rejection reason.

| Criterion | Default | Rationale |
|---|---|---|
| fit failed | - | no measurement |
| saturated | pixel ≥ 65,535 | intensity unknown |
| size (2σ) | 140-500 nm | below PSF = noise/hot pixels; far above = clusters/blobs. 140-400 nm retained 97-98% of neurite spots in a test image |
| SNR | ≥ 5 per setup; batch overrides | separates spots from noise; see below |
| intensity cutoff | off / manual per pair | dim vs single-mRNA population; never automatic |

**Setup- and batch-specific SNR.** SNR is a ratio, but false positives still differed between batches: the VisiView/EMCCD batch showed 2-3× more counted spots than the other replicates at half the median spot SNR (e.g. 2.7 vs 0.8-0.9 per 100 µm² of field; SNR 8.7 vs 20-23), consistent with amplified EMCCD noise. A third replicate showed more false positives than the second at equal SNR. Cutoffs can therefore be set per setup and overridden per pair, replicate × timepoint, replicate or timepoint, but never between control and protease of a pair. They must be chosen from QC overlays, not to equalise densities. The cutoff of every pair is written to `SNR_cutoff_per_pair.csv`.

**No automatic intensity cutoff.** On 2304 × 2304 data the bright spots form only a shoulder (~5%), so mixture models (Ashman-type separation D and dip tests) declared "one population" and would have counted the dim puncta. The script therefore only proposes a cutoff per pair; it is applied only if entered by hand.

## 8. Caching and reproducibility

Pass 1 (fits, compartment assignment, areas) is cached per image in `_pipeline_cache`. A spot file newer than its cache triggers a refit; changed mask settings (probability threshold, dilation) trigger a fast rebuild of masks and areas with the fits kept; filters are always re-applied in pass 2. Every run writes `Analysis_parameters_v4.txt`, and all downstream scripts read the same masks and filtered spots.

# 9. Quality control

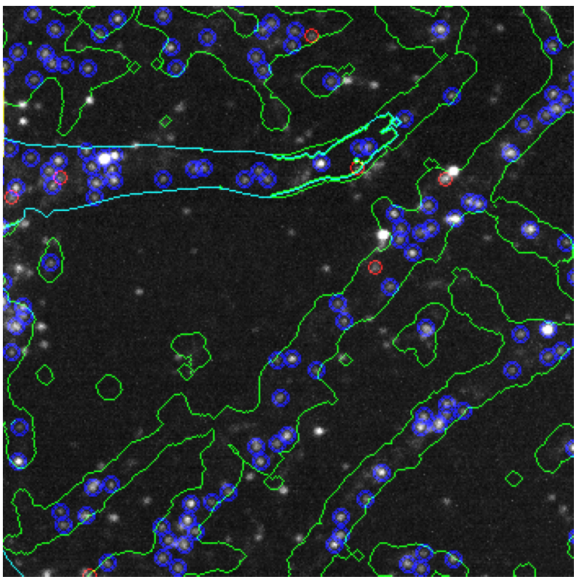


*Figure 4. QC overlay (zoom): counted spots dark-blue rings, rejected red, axon green, dendrite cyan.*

**QC overlays** are 6-channel ImageJ composites (raw, three outlines, counted, rejected) for 3 random images per folder or all. **check_overdetection.py** compares each image with the other images of its pair (count, brightness, near-duplicates < 250 nm, weak-spot fraction). In the project data it flagged 120 of 1,285 images; 26 (2%) combined dim spots with ≥ 1.5× the spot count of their condition, and in one pair the protease images had spots 0.28× as bright and 4.9× as numerous as control. It cannot detect overdetection that affects a whole pair or batch equally (e.g. the VisiView batch); batch-level comparison and overlays are needed for that.

# 10. Statistics

**Unit of replication.** Images are averaged per condition and differentiation; n = differentiations (Lord et al., 2020). Testing images as n inflates significance.

| | t-tests | Robust linear model |
|---|---|---|
| Model | per probe, on log10 replicate means | per compartment × timepoint (or condition), across all probes: mean ~ probe × group + differentiation (fixed effect); contrast of the group within each probe |
| Control vs protease | paired | as above |
| DIV9 vs DIV27 | paired where all replicates share differentiations, Welch otherwise | redone DIV9 samples coded as their own differentiation level |
| Fit | scipy | statsmodels RLM, Huber ψ (k = 1.345, as MASS::rlm default; Huber, 1964; Venables & Ripley, 2002; Seabold & Perktold, 2010); contrasts with residual df (or df = ∞) |
| Multiple testing | optional Benjamini-Hochberg (1995); off by default | same |

**Why both, and what was checked.** The robust linear model follows a published precedent from the laboratory. Simulations were used to check each variant (500 repetitions, no true effect, differentiation-specific responses):

| Variant | False-positive rate (nominal 5%) |
|---|---|
| paired t-test on replicate means | 5% |
| linear or robust model on individual IMAGES, differentiation fixed | 35-36% → rejected |

| Variant | False-positive rate (nominal 5%) |
|---|---|
| robust model on differentiation MEANS across probes (implemented) | no excess false positives in synthetic data with known effects; found 10/18 vs 6/18 real effects compared with t-tests |

The pooled model gains power by estimating variability from all probes jointly; it assumes similar variability across probes and a common differentiation effect. For a 1.5× effect with moderate consistency, a paired t-test with n = 3 has ~34% power (n = 5: ~76%). Report ratios with 95% CIs alongside p. **Area check:** compartment areas should not change with protease; ratios outside 0.8-1.25 are flagged by size, because with n = 3 a 30% area change was not significant in simulation.

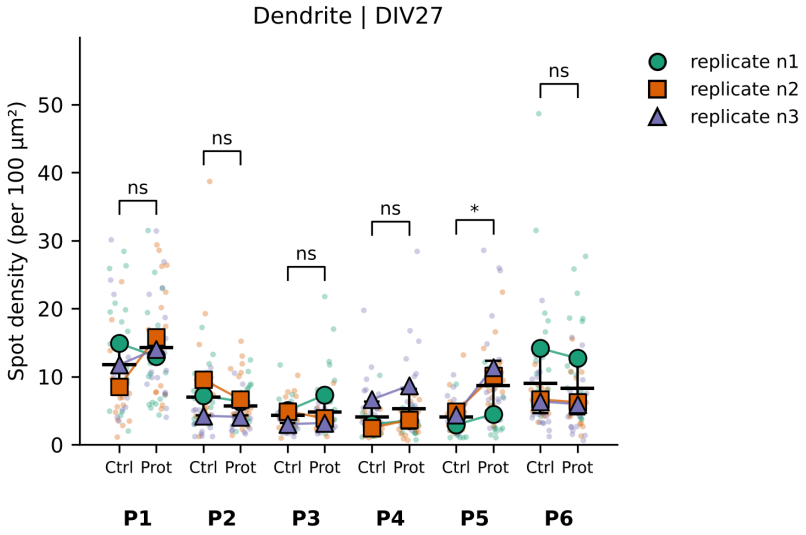



*Figure 5. SuperPlot layout (simulated data): images (small dots), replicate means (symbols), paired lines, mean ± SD, significance from the selected analysis.*

## 11. Dendrite vs axon, areas and confluence

Dendrite vs axon uses the same two analyses (paired across compartments of the same images). **Areas:** per-sample median areas are compared with the timepoint median (flag > 1.5×); a type-II ANOVA on log sample medians attributes area differences to differentiation, probe or condition. **Confluence vs density:** log10(density) ~ pair + condition + log10(axon area), robust fit, so the slope (elasticity) is estimated from confluence differences within control/protease pairs. A between-sample model produced 16-19% false positives in simulation, the within-pair model 3%. Confluence and biology of a differentiation cannot be separated with three differentiations.




*Figure 6. Per-sample axon areas (simulated data): band = ±1.5× of the timepoint median.*

# 12. Advanced quantifications

- **Distance from the soma**: geodesic path length through the dendrite mask from the soma border (skimage MCP_Geometric); never shorter than the straight distance (verified). Profiles in 5 µm bins; distal index = mean spot distance / mean dendrite-pixel distance.
- **Localization ratios** dendrite/soma and dendrite/axon per image; they cancel probe labelling efficiency.
- **mRNAs per spot**: single-mRNA unit per pair = mode of the log integrated intensity of counted neurite spots (control and protease pooled); multi-mRNA ≥ 1.5 units. Most dendritic mRNAs travel singly (Batish et al., 2012), supporting the mode as unit.
- **Morphology** of the soma-connected dendritic tree: soma area, pruned skeleton length (spurs < 3 µm removed, holes < 2 µm² filled), primary dendrites (trunks crossing the 2-5 µm ring), reach. Branch points are reported per image only: frayed tips created 17 "branch points" on a neuron with three unbranched dendrites.

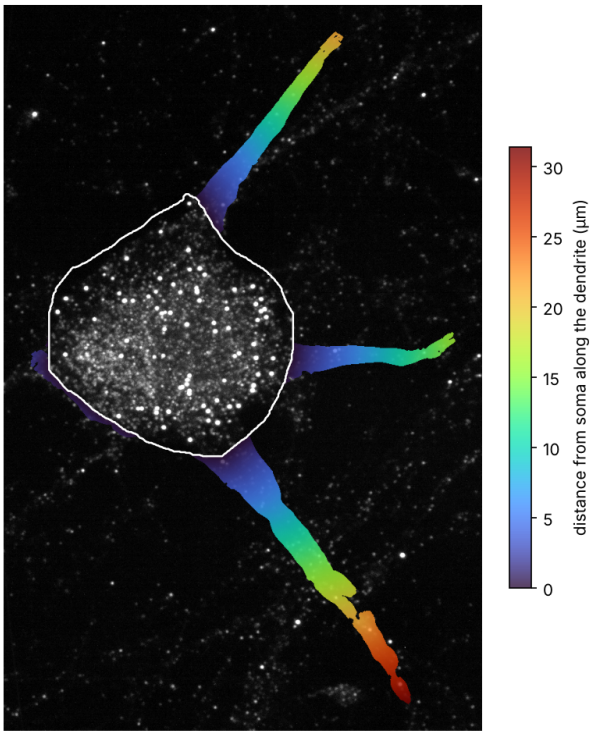



*Figure 7. Geodesic distance from the soma along the dendrites (example neuron).*

## 13. Plot editor

A Tk application that redraws SuperPlots of all analyses with the shared drawing code. Probe subsets and replicate exclusion are applied before the statistics are recomputed with the same functions; recomputed values matched the scripts to $\leq 10^{-14}$. Exclusions are annotated on exported panels and in exported statistics.

## 14. Validation summary

| Item | Check | Result |
|---|---|---|
| File pairing | IDs 1 vs 10, 001 vs 010/0010; four naming conventions | correct |
| Pixel size | metadata vs fitted PSF width, four setups | consistent; VisiView corrected |
| Background | fitted offset vs annulus, 300 spots | annulus biased (353 vs 224 counts) |
| Noise estimate | simulated noise | within ~5% |
| Per-setup / batch SNR | one setup or batch changed | only that subset changed |
| Mask dilation | ring densities; fits unchanged on mask update | as reported in §5 |
| Statistics | simulations, independent statsmodels fits | see §10; contrasts identical to formula fits |
| Confluence model | simulation | 3% vs 16-19% false positives |
| Geodesic distance | path ≥ straight distance | true for all pixels |
| End-to-end | all scripts on a demo project and simulated 6-probe data (pandas 3.0, Python 3.13) | run without errors |
| Spotiflow call | not executable in the development environment | **untested** |

## 15. Limitations and open issues

- **2D projections.** Spots overlapping in z merge (mainly soma). Collapsing a thick soma and thin neurites into one plane inflates somatic density relative to neurites; between-compartment ratios involving the soma (dendrite/soma) should be interpreted with this volume effect in mind. Comparisons within a compartment across conditions or timepoints are not affected in this way.
- **Detection consistency.** Original detections used per-image thresholds. Residual overdetection: the VisiView/EMCCD batch (stricter SNR or redetection required) and one pair with dim puncta in the protease condition.
- **Identity of dim puncta** is unresolved without a no-probe / non-targeting control.
- **Axon compartment** includes neighbouring cells; per-neuron totals are valid for soma + dendrites only.
- **Probe comparisons** of absolute densities are confounded by labelling efficiency; use ratios.
- **Protease step** is a protocol variable; its effect on detection (spot counts and brightness) is itself a result.
- **Statistical power** with n = 3 is limited; replicate exclusion should be justified technically.
- **Transformations.** The robust model uses untransformed or log10 means; a square-root option is not implemented.

## 16. Version history

| Version | Main changes |
|---|---|
| v1 | original script; fixed 0.13 µm/px; annulus background; different amplitude filters for dendrites and axons; substring file matching |
| v2 | exact ID pairing; pixel size 0.065 µm/px with check; identical filters; fitted-offset background; integrated intensity; size 140-400 nm; soma; ImageJ composite QC |
| v3 | SNR filter with noise-only simulation and threshold report; rejection reasons; four naming conventions |
| v4 (current) | batch processing of the whole project; pixel size per image; parallel fitting with cache; per-setup and per-batch SNR; µm-based mask dilation with automatic mask update; manual intensity cutoff per pair; 6-channel QC overlays; project tables; statistics with t-tests and robust model; compartments/areas, advanced quantifications, plot editor, helper scripts |

Results from v1-v3 are not comparable with v4 (pixel size, background, filters, masks).

## 17. References

*Journal articles verified in PubMed (October 2026); statistics references verified via publisher records.*


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